Sure, Your Algorithm Is Really Fast, But Is It Really Correct?

Jakob Nordström

University of Copenhagen and Lund University

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The Success of Combinatorial Solving (and the Dirty Little Secret)

- Revolution since the turn of the millennium in automated reasoning for so-called combinatorial solvers for, e.g.:
  - Boolean satisfiability (SAT) solving and optimization [BHvMW21]
  - Constraint programming [RvBW06]
  - Mixed integer linear programming [AW13, BR07]
  - Satisfiability modulo theories (SMT) solving [BHvMW21]
- Often solve these very hard problems extremely successfully in practice!
- Except the solvers are sometimes wrong... (even best commercial ones) [BLB10, CKSW13, AGJ+18, GSD19, BMN22, BBN+23, GCS23, Tin24]
- Even worse: No way of knowing for sure when errors happen

What Can Be Done About Solver Bugs?

- **Software testing**
  Very useful, but bugs slip through even with careful domain-specific testing
  Progress using fuzzing and delta debugging [BB09, BLB10, KB22, NPB22, PB23]
  But testing inherently can only detect presence of bugs, not absence

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- **Formal verification**
  Prove that solver implementation adheres to formal specification
  Current techniques cannot scale to level of complexity in modern solvers
  (Despite valiant efforts in, e.g., [Fle20])

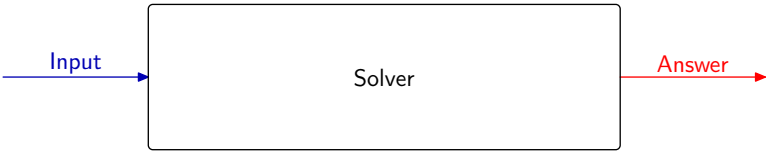
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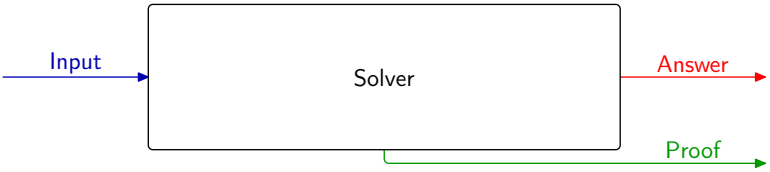
- **Proof logging**
  Make solver certifying [ABM+11, MMNS11] by adding code so that it outputs
  1. not only answer but also
  2. simple, machine-verifiable proof that answer is correct

Proof Logging with Certifying Solvers: Workflow



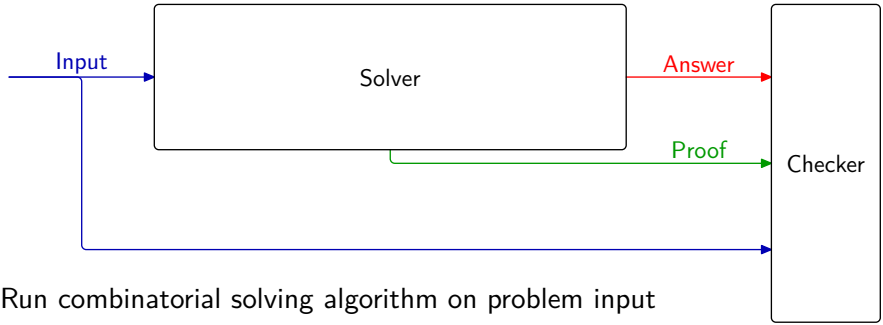
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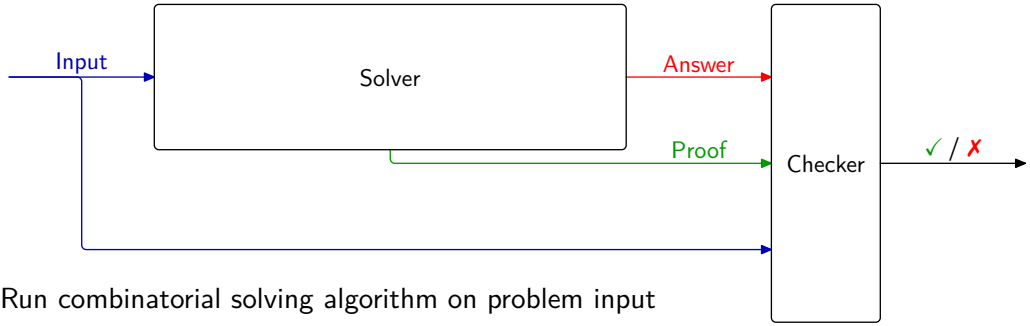
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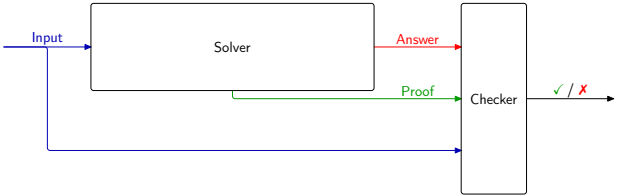
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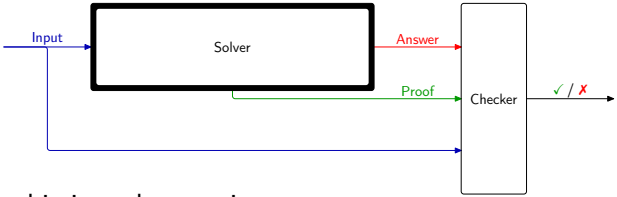
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Proof Logging Desiderata

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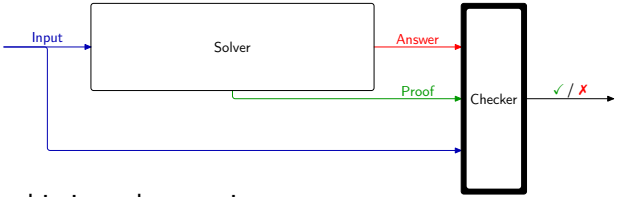
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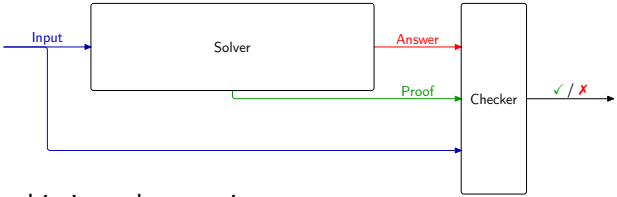
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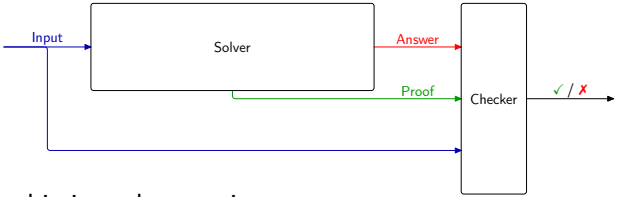


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Asking for both perhaps a little bit too good to be true?

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These slides with references are online at `https://jakobnordstrom.se/presentations/`

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7. Serves as stepping stone towards explainability
8. Offers a framework for AI agents by requiring that code should generate correct proofs

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Proof logging implementation

- Don't change solver
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Proof system

- Keep language simple — no CP constraints, SMT theories, symmetry groups, ...
- But reason efficiently about such notions using power of proof system
- Combine proof logging with formally verified proof checker

The Boolean Satisfiability (SAT) Problem

- Variable x: takes value **true** $(=1)$ or **false** $(=0)$
- Literal ℓ: variable x or its negation $\overline{x}$
- Clause $C = \ell_1 \vee \cdots \vee \ell_k$: disjunction of literals
  (Consider as sets, so no repetitions and order irrelevant)
- Conjunctive normal form (CNF) formula $F = C_1 \wedge \cdots \wedge C_m$: conjunction of clauses

The SAT Problem

Given a CNF formula F, is it satisfiable?

For instance, what about:

$$(p \vee \overline{u}) \wedge (q \vee r) \wedge (\overline{r} \vee w) \wedge (u \vee x \vee y) \wedge$$
$$(x \vee \overline{y} \vee z) \wedge (\overline{x} \vee z) \wedge (\overline{y} \vee \overline{z}) \wedge (\overline{x} \vee \overline{z}) \wedge (\overline{p} \vee \overline{u})$$

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For unsatisfiability: a proof is a sequence of clauses

- Each clause follows "obviously" from everything we know so far
- Final clause is empty, meaning contradiction (written $\bot$)
- Means original formula must be inconsistent

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Proof checker should know how to unit propagate until saturation

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DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

"Proof trace": when backtracking, write negation of decisions made

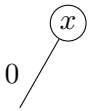
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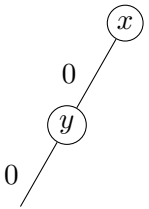


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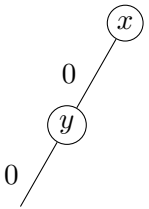


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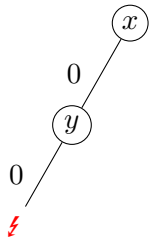
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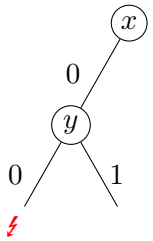
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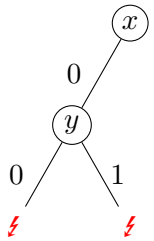
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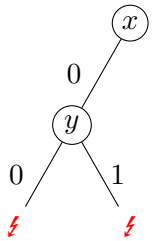
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3. x



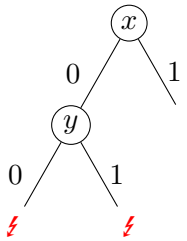
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2. $x \vee \overline{y}$
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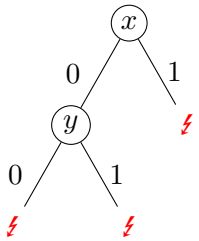
Davis-Putnam-Logemann-Loveland (DPLL) SAT Solving from the 1960s

DPLL [DP60, DLL62]: Assign variables and propagate; backtrack when clause violated

"Proof trace": when backtracking, write negation of decisions made

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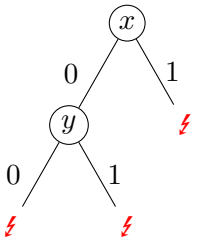
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Reverse Unit Propagation (RUP)

- Backtrack clauses are valid constraints inferred by DPLL algorithm
- To make this into a proof, need backtrack clauses to be easily verifiable

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Fact

Backtrack clauses from DPLL solver generate a proof consisting of RUP clauses

What About Modern Conflict-Driven Clause Learning (CDCL)?

Run CDCL SAT solver [BS97, MS99, MMZ+01] on our favourite CNF formula:

$$(p \vee \overline{u}) \wedge (q \vee r) \wedge (\overline{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \overline{y} \vee z) \wedge (\overline{x} \vee z) \wedge (\overline{y} \vee \overline{z}) \wedge (\overline{x} \vee \overline{z}) \wedge (\overline{p} \vee \overline{u})$$

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Forced choice to avoid falsifying clause

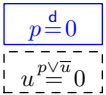
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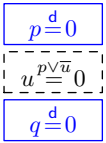
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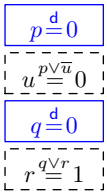
Add to assignment trail

Continue until satisfying assignment or conflict

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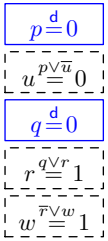
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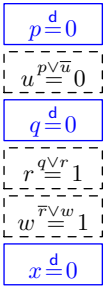
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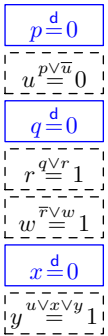
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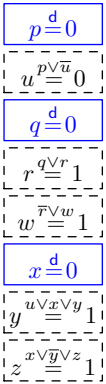
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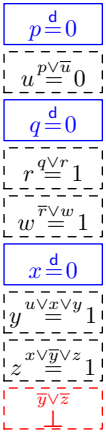
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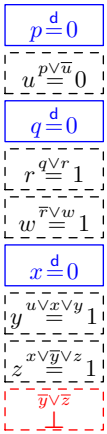
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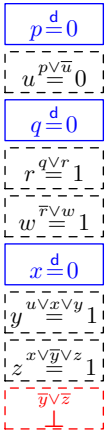
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Conflict Analysis

Time to analyse this conflict and learn from it!

$$(p \vee \overline{u}) \wedge (q \vee r) \wedge (\overline{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \overline{y} \vee z) \wedge (\overline{x} \vee z) \wedge (\overline{y} \vee \overline{z}) \wedge (\overline{x} \vee \overline{z}) \wedge (\overline{p} \vee \overline{u})$$



decision level 1

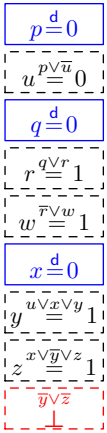
decision level 2

decision level 3

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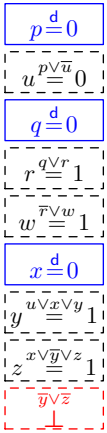


Could backtrack by erasing conflict level & flipping last decision

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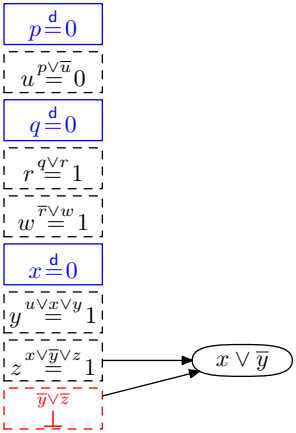
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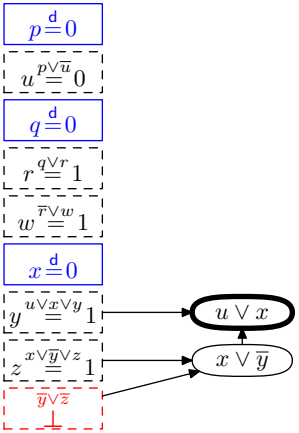
Case analysis over z for last two clauses:

- $x \vee \overline{y} \vee z$ wants $z = 1$
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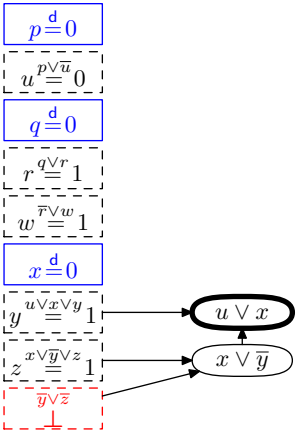
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Repeat until UIP clause with only 1 variable at conflict level after last decision — learn and backjump

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

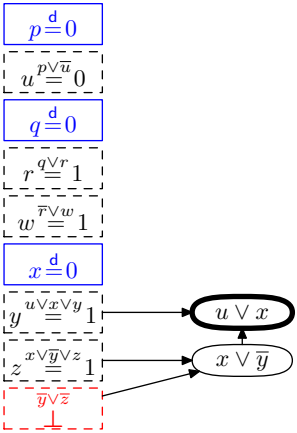
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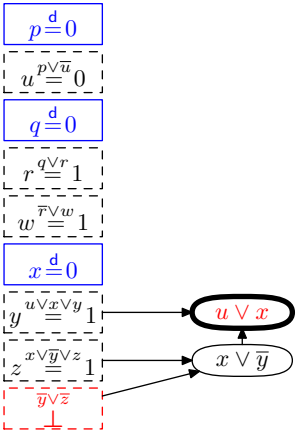


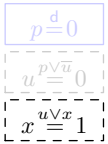
Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

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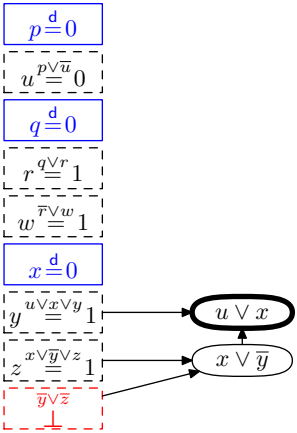
Assertion level 1 (2nd largest level in learned clause) — trim trail to that level

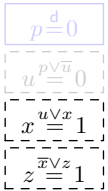
Now UIP literal guaranteed to flip (assert) — but this is a propagation, not a decision

Complete Example of CDCL Execution

Backjump: undo max #decisions while learned clause propagates

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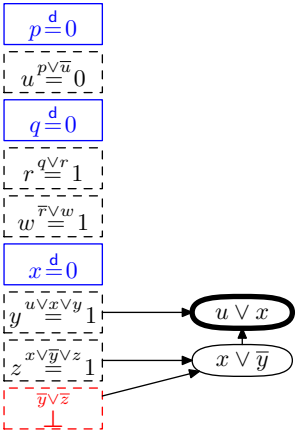
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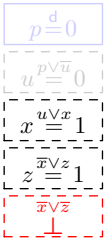
Then continue as before...

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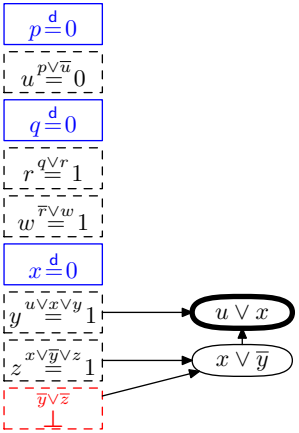


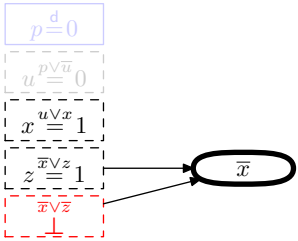


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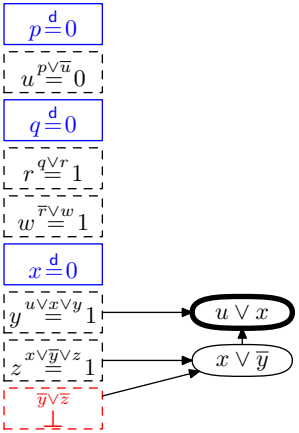


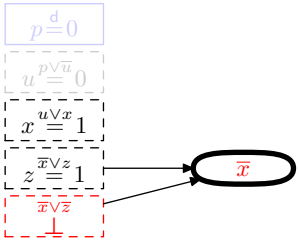


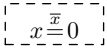
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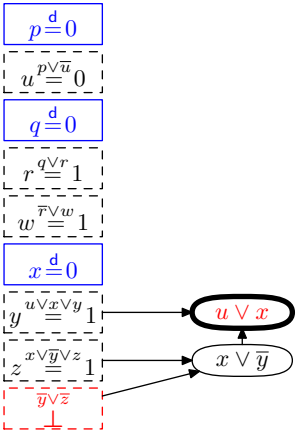


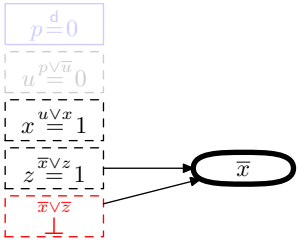


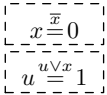
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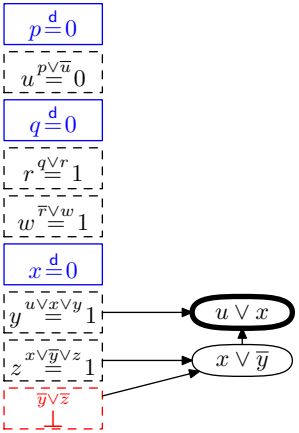


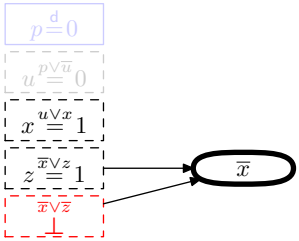


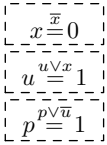
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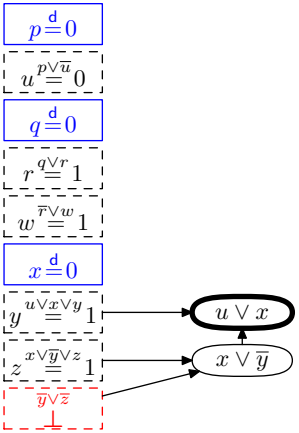


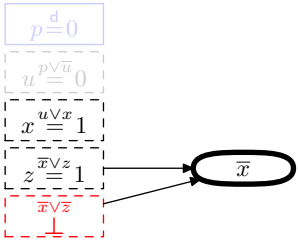


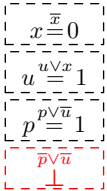
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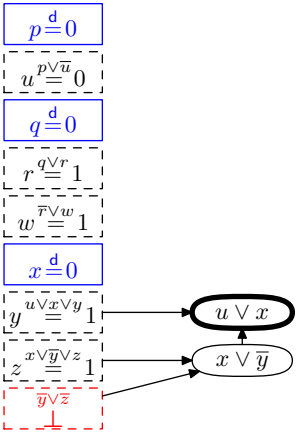


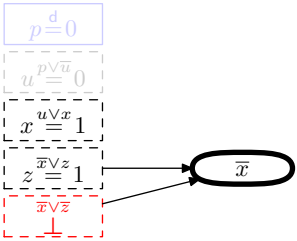


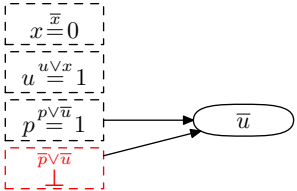
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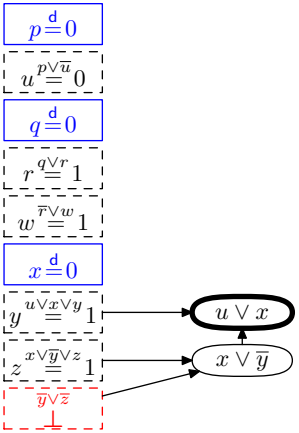


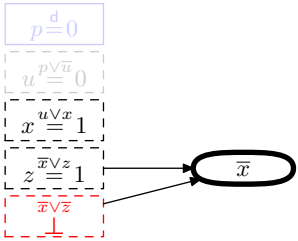


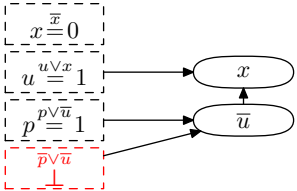
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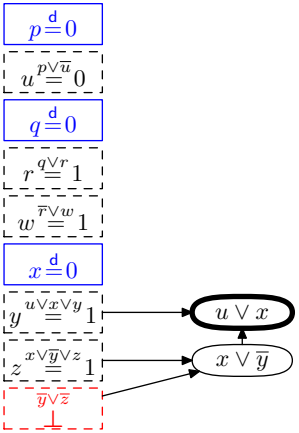


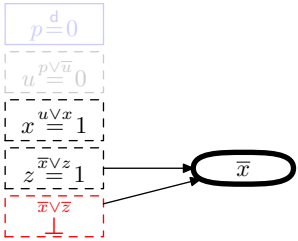


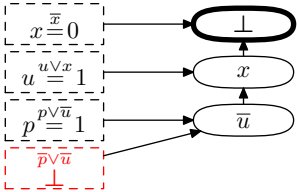
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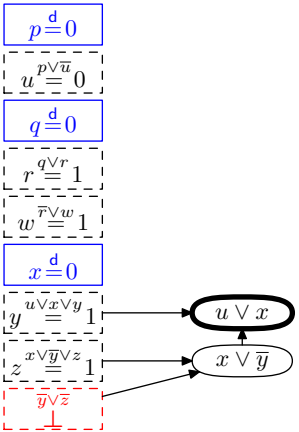
(*) Ignores pre- and inprocessing, but we don't have time to go into this level of detail

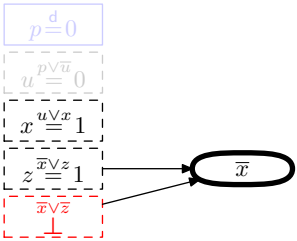
Resolution Proofs from CDCL Executions

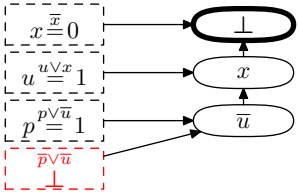
Obtain resolution proof...

Resolution Proofs from CDCL Executions

Obtain resolution proof from our example CDCL execution...

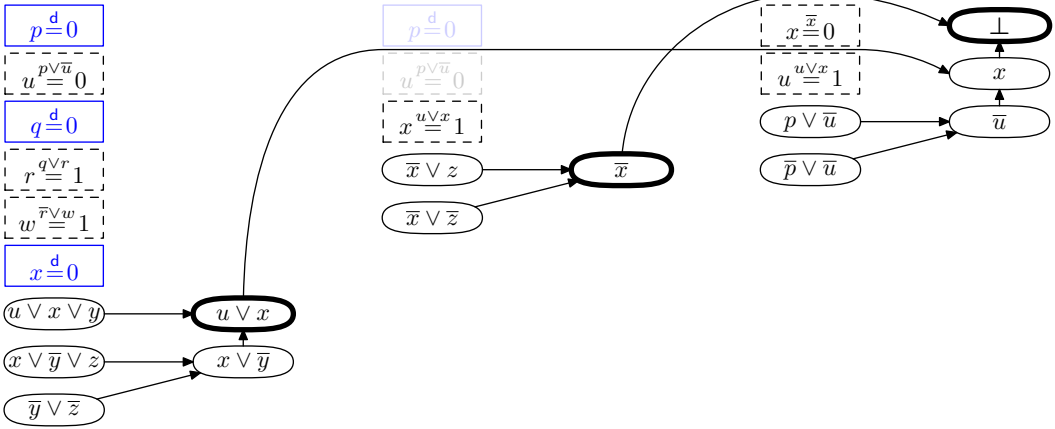






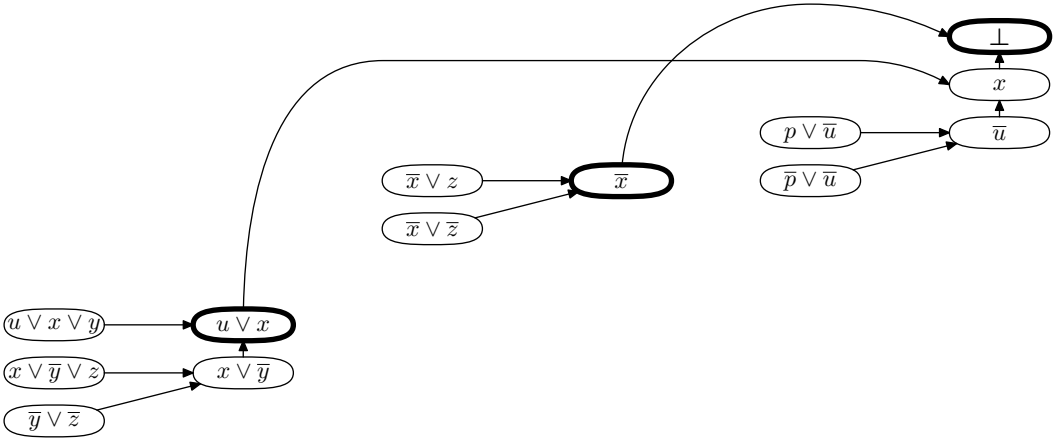
Resolution Proofs from CDCL Executions

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RUP Proofs and CDCL

But it turns out we can be lazier...

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All learned clauses generated by CDCL solver are RUP clauses

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RUP Proofs and CDCL

But it turns out we can be lazier...

Fact

All learned clauses generated by CDCL solver are RUP clauses

So a shorter short proof of unsatisfiability for

$$(p \vee \overline{u}) \wedge (q \vee r) \wedge (\overline{r} \vee w) \wedge (u \vee x \vee y) \wedge (x \vee \overline{y} \vee z) \wedge (\overline{x} \vee z) \wedge (\overline{y} \vee \overline{z}) \wedge (\overline{x} \vee \overline{z}) \wedge (\overline{p} \vee \overline{u})$$

is the sequence of reverse unit propagation (RUP) clauses

1. $u \vee x$
2. $\overline{x}$
3. $\bot$

Why Aren't We Done?

Practical limitations of current SAT proof logging technology:

- Difficulties dealing with stronger reasoning efficiently (even for SAT solving)
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- 0–1 linear inequalities $\sum_i a_i \ell_i \geq A$
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Use pseudo-Boolean reasoning with (extension of) the cutting planes proof system [CCT87]

Successful Applications of Pseudo-Boolean Proof Logging with VeriPB

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Surprisingly, pseudo-Boolean reasoning is sufficient to efficiently certify a wide range of combinatorial solving techniques and paradigms:

1. Boolean satisfiability (SAT) solving including advanced techniques such as
   - Gaussian elimination [GN21]
   - symmetry breaking [BGMN23, ABB+26b]
2. SAT-based optimization (MaxSAT) [VDB22, BBN+23, BBN+24, IOT+24, VCB26]
3. Pseudo-Boolean solving and optimization [GMNO22, KLM+25]
4. Graph solving [GMN20, GMM+20, GMM+24, DKK+26]
5. Dynamic programming and decision diagrams [DMM+24]
6. Presolving in 0–1 integer linear programming [HOGN24]
7. Constraint programming [EGMN20, GMN22, MM23, MMN24, MM25]
8. Automated planning [DHN+25]
9. Enumeration of solutions/models [MNOT26]

Future Challenges

User-friendliness

- Make proof logging easier for solver authors
- Make proof checking faster
- Improve support beyond $\{0, 1\}$-valued problems for, e.g.,
  - Mixed integer linear programming *(suggested extension of* VeriPB *in [DEGH23])*
  - Satisfiability modulo theories (SMT) solving *(lots of work in, e.g., [HS22, BBC+23, RSB+26])*

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Explainability, robustness, fairness

- Certified explanations for decisions or classifications?
- Proofs of neural network robustness to limited input perturbations?
- Rigorous symbolic reasoning about fairness?

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Lots of other challenging problems and interesting ideas

- This talk will probably sound quite different in two-three years from now

VeriPB Resources

VeriPB tutorials, technical documentation, and more information at https://VeriPB.org



Tutorial series with videos and slides from *WHOOPS '25* workshop at https://jakobnordstrom.se/WHOOPS25/

Technical details in papers [EGMN20, GMN20, GMM+20, GN21, GMN22, GMNO22, VDB22, BBN+23, BGMN23, MM23, BBN+24, DMM+24, GMM+24, HOGN24, IOT+24, MMN24, DHN+25, JBBJ25, KLM+25, MM25, ABB+26a, ABB+26b, DKK+26, MNOT26, VCB26] and PhD theses [Goc22, McI26, Oer26]

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We're happy to collaborate — please get in touch if you have computationally challenging problems where correctness is absolutely crucial!

And we're hiring — talk to me if you want to join the proof logging revolution! ☺

Summing up

- Automated reasoning and combinatorial solving are true success stories
- But ensuring correctness is a crucial, and not yet satisfactorily addressed, concern
- Certifying solvers producing machine-verifiable proofs of correctness is the most promising approach
- Cutting planes reasoning with pseudo-Boolean constraints seems to hit a sweet spot between simplicity and expressivity
- **Action point:** What problems can VeriPB solve for you? ☺



Summing up

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- **Action point:** What problems can VeriPB solve for you? ☺

Thank you for your attention!



Some Types of Pseudo-Boolean Constraints

1. Clauses

$$x_1 \vee \overline{x}_2 \vee x_3 \quad \Leftrightarrow \quad x_1 + \overline{x}_2 + x_3 \geq 1$$

2. Cardinality constraints

$$x_1 + x_2 + x_3 + x_4 \geq 2$$

3. General pseudo-Boolean constraints

$$x_1 + 2\overline{x}_2 + 3x_3 + 4\overline{x}_4 + 5x_5 \geq 7$$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms From the input

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms From the input

Literal axioms $\dfrac{}{\ell_i \geq 0}$

Pseudo-Boolean Reasoning: Cutting Planes [CCT87]

Input/model axioms From the input

Literal axioms

$$\frac{}{\ell_i \geq 0}$$

Addition

$$\frac{\sum_i a_i \ell_i \geq A \qquad \sum_i b_i \ell_i \geq B}{\sum_i (a_i + b_i)\ell_i \geq A + B}$$

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Multiplication for any $c \in \mathbb{N}^+$

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i c a_i \ell_i \geq cA}$$

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Division for any $c \in \mathbb{N}^+$
(assumes normalized form)

$$\frac{\sum_i a_i \ell_i \geq A}{\sum_i \left\lceil \frac{a_i}{c} \right\rceil \ell_i \geq \left\lceil \frac{A}{c} \right\rceil}$$

Cutting Planes Toy Example

$$w + 2x + y \geq 2$$

Cutting Planes Toy Example

Multiply by 2 $\dfrac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$

Cutting Planes Toy Example

Multiply by 2 $\dfrac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}$ $\quad w + 2x + 4y + 2z \geq 5$

Cutting Planes Toy Example

$$\text{Multiply by } 2 \;\; \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} \qquad \text{Add} \;\; \frac{2w + 4x + 2y \geq 4 \qquad w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9}$$

Cutting Planes Toy Example

$$\text{Multiply by } 2 \quad \frac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4} \qquad \text{Add} \quad \frac{w + 2x + 4y + 2z \geq 5 \qquad \overline{z} \geq 0}{3w + 6x + 6y + 2z \geq 9}$$

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$$\dfrac{\dfrac{w + 2x + y \geq 2}{2w + 4x + 2y \geq 4}\ \text{Multiply by 2} \qquad w + 2x + 4y + 2z \geq 5}{3w + 6x + 6y + 2z \geq 9}\ \text{Add} \qquad \dfrac{\overline{z} \geq 0}{2\overline{z} \geq 0}\ \text{Multiply by 2}$$

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Cutting Planes Toy Example

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By referring to constraints by labels and to literal axioms by the literal involved as

$$\begin{aligned} \texttt{@C1} &\doteq 2x+y+w\geq 2 \\ \texttt{@C2} &\doteq 2x+4y+2z+w\geq 5 \\ \sim\texttt{z} &\doteq \overline{z}\geq 0 \end{aligned}$$

Cutting Planes Toy Example

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such a calculation is written in the proof log in reverse Polish notation as

```
pol  @C1  2  *  @C2  +  ~z  2  *  +  3  d  ;
```

Resolution and Cutting Planes

To simulate resolution step such as

$$\frac{\overline{y} \vee \overline{z} \qquad x \vee \overline{y} \vee z}{x \vee \overline{y}}$$

we can perform the cutting planes steps

$$\begin{array}{l} \text{Divide by } 2 \quad \dfrac{x + 2\overline{y} \geq 1}{x + \overline{y} \geq 1} \\ \text{Add} \quad \dfrac{\overline{y} + \overline{z} \geq 1 \qquad x + \overline{y} + z \geq 1}{x + 2\overline{y} \geq 1} \end{array}$$

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we can perform the cutting planes steps

$$\begin{array}{l} \text{Add} \quad \dfrac{\overline{y} + \overline{z} \geq 1 \qquad x + \overline{y} + z \geq 1}{x + 2\overline{y} \geq 1} \\ \text{Divide by } 2 \quad \dfrac{x + 2\overline{y} \geq 1}{x + \overline{y} \geq 1} \end{array}$$

Given that the premises are clauses 7 and 5 in our example CNF formula, using references

$$\begin{aligned} \texttt{@C7} &\doteq \overline{y} + \overline{z} \geq 1 \\ \texttt{@C5} &\doteq x + \overline{y} + z \geq 1 \end{aligned}$$

we can write this in the proof log as

```
pol  @C7  @C5  + 2  d
```

Pseudo-Boolean Proof Logging — How and Why?

If problem is (special case of) 0-1 integer linear program

- just do proof logging [basically: add print statements to solver code]

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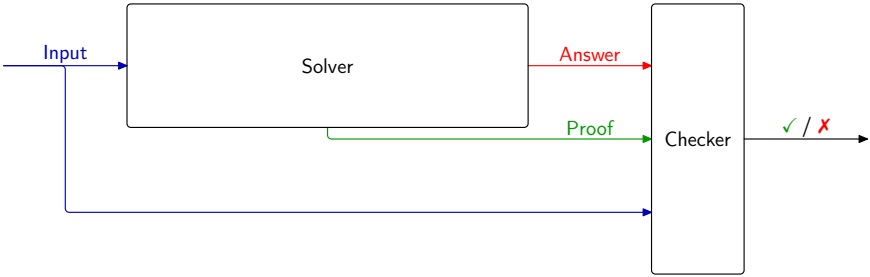
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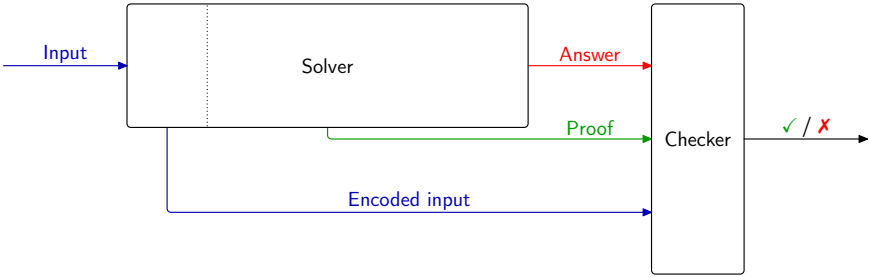
Goldilocks compromise between expressivity and simplicity:

1. 0-1 ILP expressive formalism for combinatorial problems (including objective)
2. Powerful reasoning capturing many combinatorial arguments

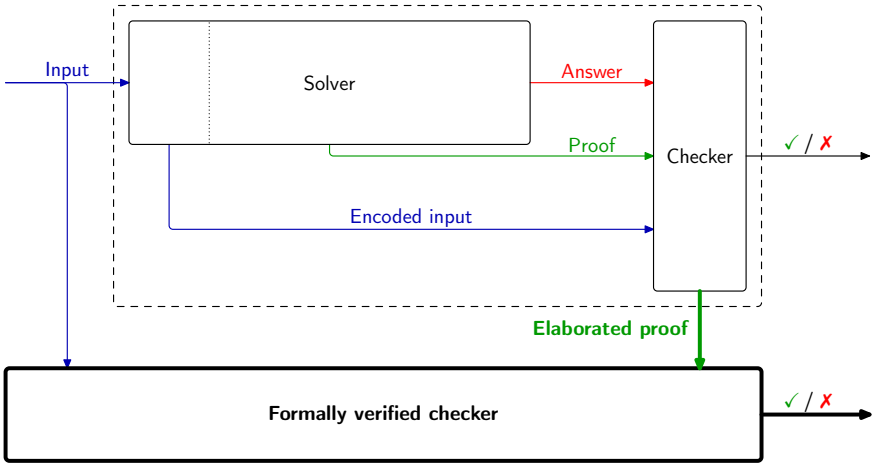
Proof Logging with Formally Verified Checking: Full Workflow



Proof Logging with Formally Verified Checking: Full Workflow



Proof Logging with Formally Verified Checking: Full Workflow



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